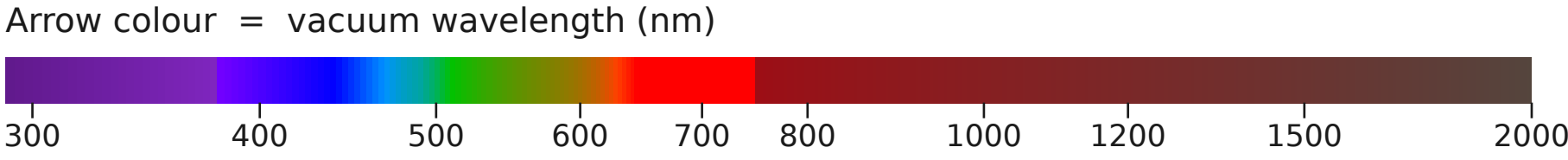


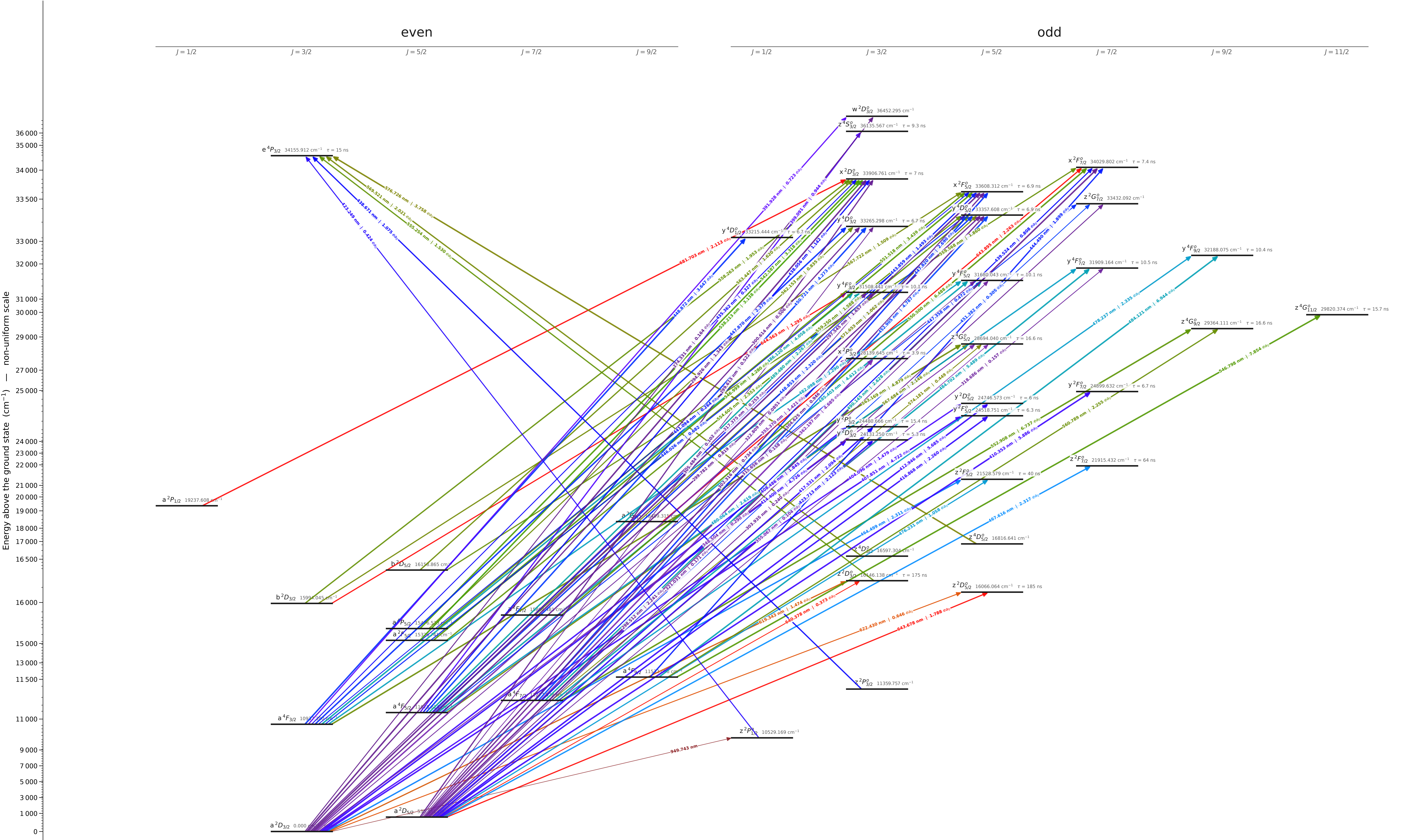
# Yttrium

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J || e r || J' \rangle$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Y (Y I). Strongest classified transitions below 2  $\mu\text{m}$  (NIST): 44 levels, 90 lines.

Y<sup>+</sup> ionisation limit 50145.60  $\text{cm}^{-1}$  = 6.21726 eV (199.419 nm)



## Key transitions

| Transition                   | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ \langle J    e r    J' \rangle $ ( $e a_0$ ) | A ( $\text{s}^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                         |
|------------------------------|--------------------|--------------------|-------------|------------------------------------------------|-----------------------|-------------------------|------------|-------------------------------------------------------------|
| $a^2D_{3/2} - y^2F_{5/2}^o$  | 407.8511           | 407.7360           | 735.0537    | 4.722                                          | 1.110e+08             | 25.26                   | 70         | strong 2D-2F resonance line                                 |
| $a^2D_{5/2} - y^2F_{7/2}^o$  | 410.3527           | 410.2369           | 730.5727    | 5.886                                          | 1.270e+08             | 23.75                   | 85         | strong 2D-2F resonance line                                 |
| $a^2D_{5/2} - y^2D_{5/2}^o$  | 412.9463           | 412.8299           | 725.9841    | 5.685                                          | 1.550e+08             | 26.53                   | 93.3       | strong 2D-2D resonance line of the ground term              |
| $a^2D_{3/2} - y^2D_{3/2}^o$  | 414.4004           | 414.2836           | 723.4367    | 4.726                                          | 1.590e+08             | 30.03                   | 84.1       | strong 2D-2D resonance line of the ground term              |
| $a^4F_{9/2} - z^4G_{11/2}^o$ | 546.7983           | 546.6464           | 548.2688    | 7.854                                          | 6.370e+07             | 10.14                   | 100        | proposed laser-cooling                                      |
| $a^2D_{3/2} - z^2P_{1/2}^o$  | 949.7426           | 949.4821           | 315.6565    | -                                              | -                     | -                       | -          | proposed type-II MOT / narrow cooling line from the true... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## <sup>89</sup>Y hyperfine structure ( $I = 1/2$ )

| Level          | A (MHz)  | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|----------------|----------|---------|--------------------------------------------------------|
| $a^2D_{3/2}$   | -57.220  | -       | F=1: +71.53 F=2: -42.91                                |
| $a^2D_{5/2}$   | -28.850  | -       | F=2: +50.49 F=3: -36.06                                |
| $z^2D_{5/2}^o$ | -216.100 | -       | F=2: +378.18 F=3: -270.12                              |
| $z^2D_{3/2}^o$ | 88.300   | -       | F=1: -110.38 F=2: +66.22                               |
| $z^4D_{3/2}^o$ | -38.300  | -       | F=1: +47.88 F=2: -28.73                                |
| $z^4D_{5/2}^o$ | -132.600 | -       | F=2: +232.05 F=3: -165.75                              |
| $e^4P_{3/2}$   | -124.800 | -       | F=1: +156.00 F=2: -93.60                               |

Measured A, B constants; sources in the data file.